High-Capacity Transmission with Single External Laser Source and Polymer-Based Splitters for Co-Packaged Optics

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Abstract: We demonstrated high-capacity optical transmission using a polymer-based 1×4 splitter and a single external laser source (ELS) with an extremely high power of +20 dBm per channel, potentially achieving a bandwidth of 3.2 Tbps.

1. Introduction

Co-packaged optics (CPO) technology, which integrates photonics and electronics onto a single platform, is essential for achieving high-capacity data transmission with low power consumption in data centers and high-performance computing systems. In CPO, the optical interconnect technology between optical transceiver chip based on silicon (Si) photonics and optical fibers is of critical importance. Currently, direct coupling from Si waveguides to optical fibers is being researched [1], however, this direct coupling presents challenges due to high coupling losses and a complex assembly process. We propose an “Active Optical Package (AOP) Substrate” to connect embedded Si photonic integrated circuit (PIC) and optical fibers using three-dimensional mirrors, linking them with polymer waveguides [2]. The polymer waveguide offers better compatibility with organic substrate, low propagation loss, reduced costs, and simplified processing, presenting a significant advantage in CPO.

In recent standardization efforts, there have been proposals to use an external light source (ELS) with high optical power for distributing light to multiple modulators in a CPO system. This approach, including direct optical branching on a photonic chip, has been gaining attention [3, 4]. Currently, some ELS can output +20 dBm per channel and support CWDM4 wavelengths [5]. However, in Si waveguides, such high-power inputs often lead to nonlinear absorption issues [6], and there is a risk of waveguide damage at + 20 dBm [7]. We have previously demonstrated high-power stability in the polymer waveguides with ELS inputs over +20 dBm without excess loss [8]. This advancement of AOP substrate, illustrated in Fig. 1, proposes a practical and robust design for high-power optical transmission using ELS and polymer-based splitters for CPO. There are reports of polymer-based splitters on substrates such as Si and glass for optical devices in the C-band [9]. However, the performance of polymer optical waveguides formed on organic substrates under high-power input in the O-band remains unexplored. In this paper, we demonstrated high-capacity optical transmission using polymer-based 1×4 splitter on a glass epoxy substrate and a single ELS with a view toward a CPO system.

2. Experiment

The 1×4 Y-branch splitters based on polymer waveguide are fabricated using direct laser lithography on the glass epoxy substrate. The length of the splitter is 10 mm and the pitch of the output ports of the splitter was designed to be 62.5 μm to facilitate miniaturization of the Si photonics chip. Fig. 2 shows the insertion losses of each output port for the fabricated polymer-based 1×4 splitter for each wavelength of CWDM4 using the tunable laser source with 0 dBm input. This excess loss is assumed to be due to fabrication errors. The prototype of ELS and its evaluation board were generously provided by Furukawa Electric as shown in Fig.3(a). The ELS has 8 output channels, with each wavelength compliant with CWDM4 being output from 4 of 8 channels. The remaining 4 channels operate in the same manner. Fiber-coupled optical power is > +20 dBm per channel. In this experiment, we switched and measured each of the 4 out of 8 channels. Fig. 3(b) shows the experimental setup for high-capacity optical transmission using the splitter and the ELS. We emitted each wavelength of CWDM4 from the ELS and passed it

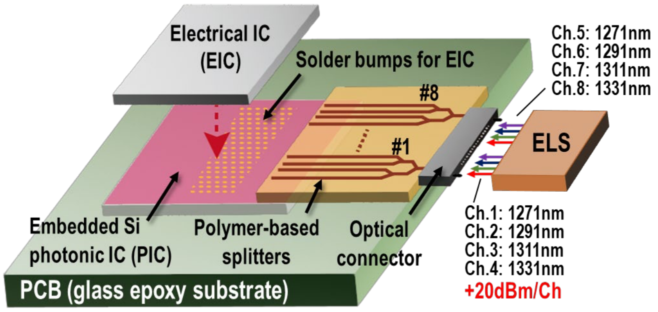

Fig. 1 Concept of “Active optical package substrate” on the input side, using an ELS and a 1×N splitters based on polymer waveguide for CPO.

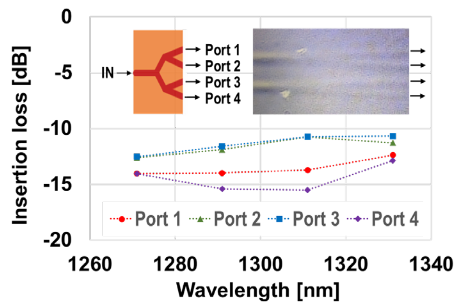

Fig. 2 Insertion losses of each output port for fabricated polymer-based 1×4 Y-branch splitter in CWDM4.

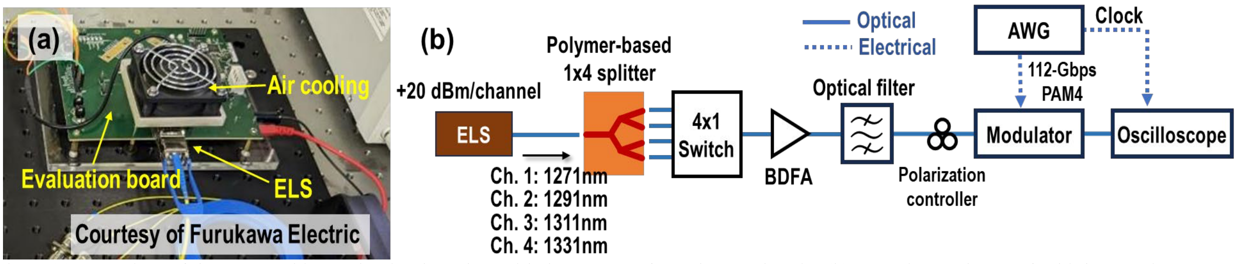

Fig. 3 (a) Overview of the ELS and its evaluation board provided courtesy of Furukawa Electric. (b) Experimental setup for high-capacity transmission using the fabricated polymer-based 1×4 splitter and the single ELS.

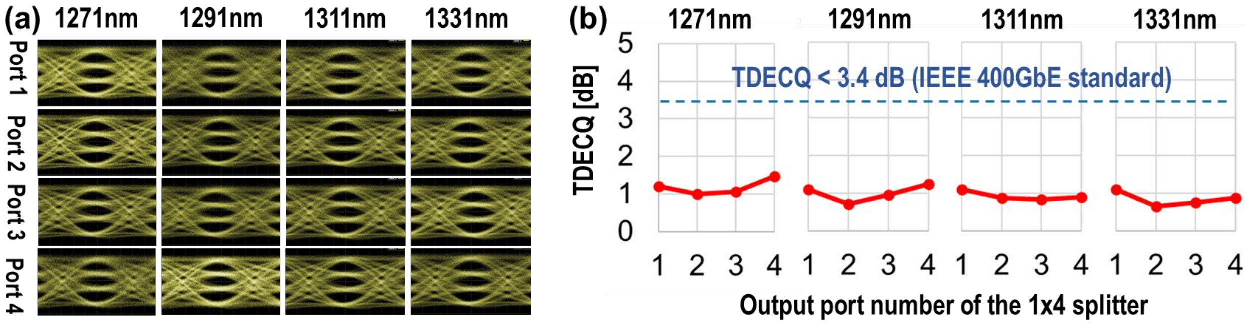

Fig. 4 (a) Modulated 112 Gbps PAM4 eye diagrams and (b) their TDECQ values from the ELS with +20 dBm per channel through the 1×4 splitter for each output port of the splitter at each wavelength of CWDM4.

through the splitter. A bismuth-doped fiber amplifier (BDFA) amplifies the optical signal to compensate for splitter losses. We note that if the loss of the splitter is improved, a BDFA is not necessary. An optical filter removes the amplified spontaneous emission (ASE) noise from the BDFA. Next, we performed 112 Gbps PAM4 modulation by an arbitrary waveform generator (AWG) and a lithium niobite intensity modulator. Finally, a digital sampling oscilloscope monitors the eye diagrams of the output signals. In this measurement, the signal quality was evaluated by the transmitter and dispersion eye closure quaternary (TDECQ). Fig. 4 shows the modulated 112 Gb/s PAM4 eye diagrams and their TDECQ values. Here, we confirmed that even when +20 dBm is input, the loss characteristic shows almost no change compared to Fig. 2. The relatively high TDECQ values of the eye diagrams at 1271 nm is due to the bandwidth dependence of gain in the BDFA. Based on these results, we have obtained the measured TDECQ values that meet the IEEE 400G Ethernet standard, which requires the TDECQ < 3.4 dB. We found that the combination of an ELS with a polymer-based splitter is effective for a high-capacity transmission for CPO.

3. Conclusion

In this experiment, we demonstrated 400 Gbps (112 Gbps × 4) transmission using one channel of a single ELS with a polymer-based 1×4 splitter, and potentially achieved 1.6 Tbps by employing 4 channels of ELS. Furthermore, with the remaining four channels of the same specification, the possibility extends to 3.2 Tbps transmission for all 8 channels of ELS. Reducing the losses in polymer-based splitters by fabrication improvement will enable the use of more branches, thereby allowing for even larger capacity in optical transmission. These results suggest that the polymer-based splitter could be one of the solutions for CPO with ELS.

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